BigFix Master Class 2026

Automated BigFix Performance and Configuration Analysis

Student Lab Guide

October 13, 2026



For more information

To learn more about BigFix, contact your HCLSoftware representative, HCL Business Partner, or visit www.BigFix.com.

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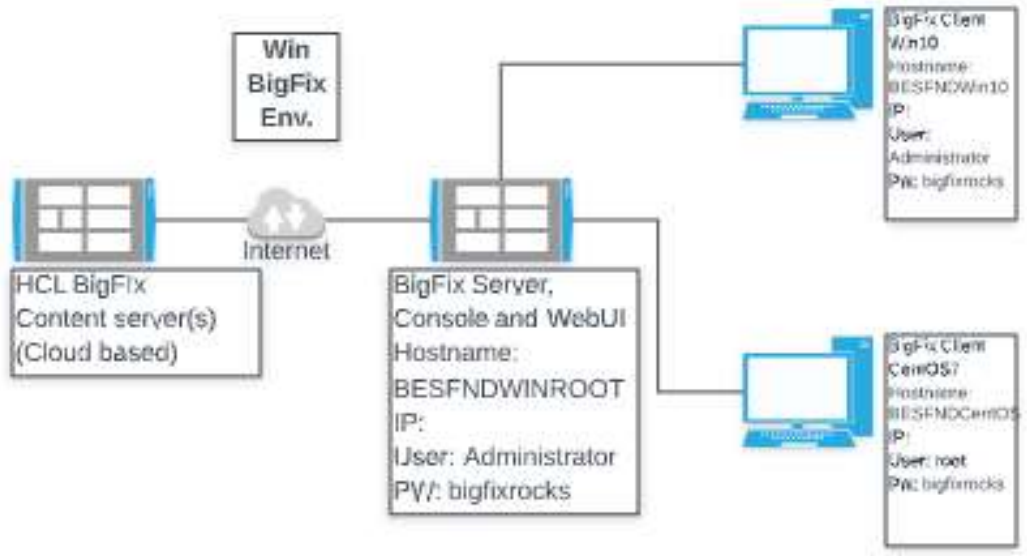
Version Control

Release Date	Version	Authors	Comments
October 7, 2026	1.0	Duncan McCalla	First version.

Lab Environment

The below table contains a summary of the VM images used in this lab guide:

Host Name	BigFix Components	OS	IP Address	Userid & Password
BESFNDWINROOT	BigFix Windows based Server, Console, WebUI, and Client	Windows 2016	10.0.0.1	Administrator bigfixrocks
BESFNDWIN10	BigFix Client, Console	Windows 10	10.0.0.2	tecuser bigfixrocks
BESFNDCENTOS	BigFix Client	CentOS7	10.0.0.3	root bigfixrocks
All	BigFix Console creds			adminmo B1gfixrocks



Accessing Lab Environment

The BigFix Lab environment is currently being hosted in Skytap's (www.skytap.com). To access this environment, you will need the url, id, and password sent to your registered email address (this would be from Skytap.com). If you are a USA Federal customer – your instructor will provide you your credentials and access url(s).

Students will receive an email (this is the email address you provided when you registered for the course) from Skytap that contains the url to YOUR Skytap environment, the login id and password for this specific course. It will look something like this:

[CAUTION: This Email is from outside the Organization. Unless you trust the sender, Don't click links or open attachments as it may be a Phishing email, which can steal your Information and compromise your Computer.]

Hello james.leaphart@hcl.com,

Event: MARK 0

Course: TEST5 US

Start time:

End time: 05/15/2020 12:34 PM PDT

Student Region: US-Central

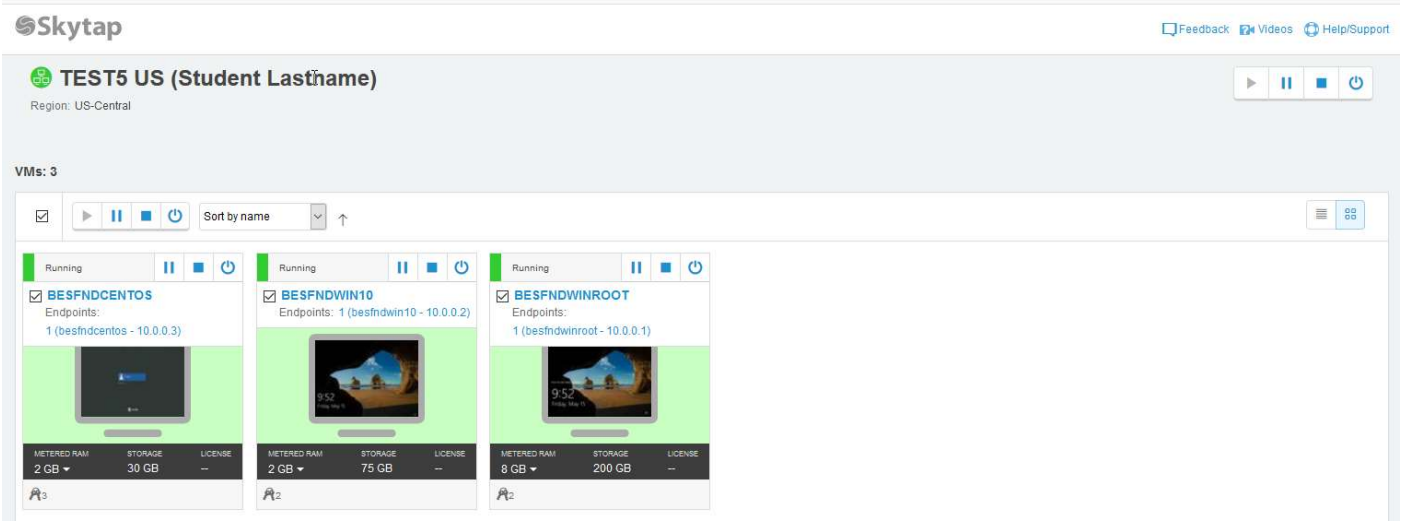
Student Passcode: P9G6ZB7APZYQ

Student URL: https://hcl-vt.skytap-portal.com/lab_access/event_participant/13/995d8455a0ac743edb1a1c6ebca90d9cc8e6805383edc11cf581887b12ceff5a

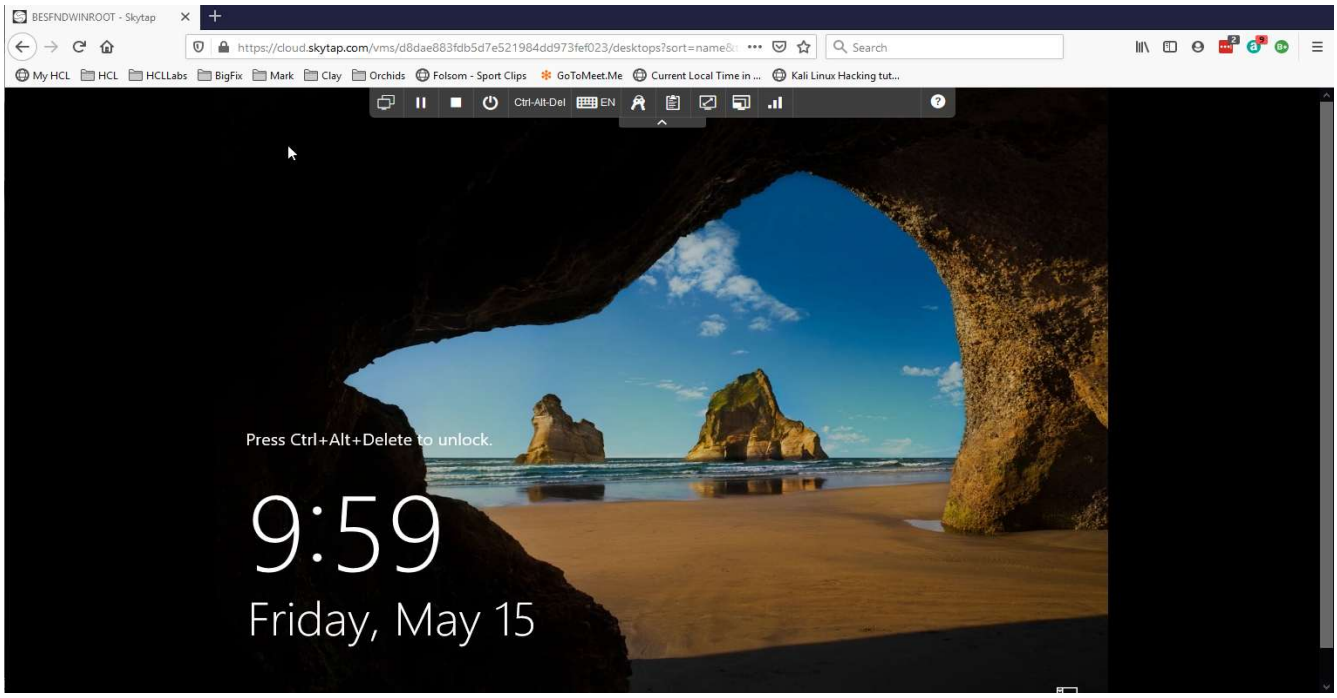
Instructors:

Instructor Email/ID	Instructor Name	Region
leaphartmark@gmail.com	Mark Leaphart	US-Central

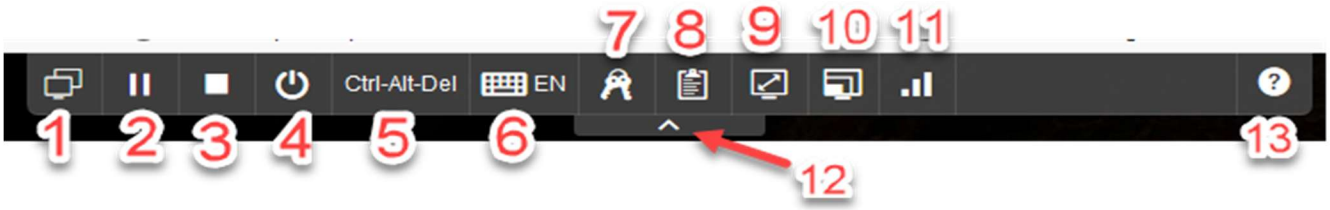
Click on the url provided in your email and provide your credentials (if asked). You will be taken into Skytap and you will see your provisioned environment.



The vm's provided here are accessible via your browser (rdp is not required). Click on a vm and your browser will present your vm:



Now let's look at the controls in the browser for this vm.



1) Environment VM's: View all vm's in your environment or switch to another vm in your environment
2) Suspend this vm
3) Shutdown this vm
4) Power options for vm - a) shutdown, b) reset, c) power off
5) Ctrl-Alt-Del is passed to the vm
6) Keyboard layout and or inject key combinations
7) Credentials: operating system and applications in this vm
8) VM Clipboard
9) Fit to window
10) Change video resolution
11) Network Quality Indicator
12) Hide this tool bar
13) Help

When you open any of the Windows vm's, always answer YES to the network connection question.

Master Class Diagnostics Workbench Lab Exercises

Overview

In this lab you will run the Master Class Diagnostics Workbench. You'll cover the two modules from the session today. The FillDB module is considered to be the Bonus Round and can be done if you finish the first two modules early OR you can do them at home/work later.

- Perfmon Analysis reviews the Windows performance of BigFix servers.
    - This can be any BigFix server (Web Reports, WebUI, etc.) and the SQL server(s) supporting the Platform. Although the focus, and established Perfmon thresholds, are typically on the Root Server and SQL Server.
    - The Workbench Perfmon Analysis module allows for the optional analysis of TWO Perfmon files in a single pass. Many customers will have a Perfmon file from their root server and another one from a SQL server.
- Client Performance Analysis reviews how quickly the BigFix clients evaluate content, respond to notifications, and how they are configured for performance.

The lab takes roughly 20 minutes…

Part	Time
Before You Start	2 minutes
Exercise 1: Collect the Data	5 minutes
Exercise 2: Run the Workbench	5 minutes
Exercise 3: Read the Client Performance Output	4 minutes
Exercise 4: Read the Perfmon Analysis Output	4 minutes

NOTE: The FillDB exercise is waaaaay back in Appendix A. It should only take about 5 minutes if you choose to do it.

Before You Start

In production environments, the collection analysis needs several hours to catch as many clients as possible, the client tasks can take a while to run too. Perfmon is also a 24-hour capture.

In this lab you look at the analysis and run the tasks only to see what they do and how they work. The Workbench then builds from some lab sample data that ships with it.

Every module follows the same basic flow:

BigFix deployment → Collection task → Captured data→ Workbench module → Mini Report

The Workbench is on the BigFix Server virtual machine in**:**

C:\Users\Administrator\Desktop\BigFixSrc\Master Class Diagnostics Workbench

The sample data is in**:**

C:\Users\Administrator\Desktop\BigFixSrc\Master Class Diagnostics Workbench\SampleData\BigFix Master Class 2026\Results

NOTE: The Workbench is provided as-is to BigFix Master Class attendees only. It has no warranty or support. Do not redistribute it without approval.

Exercise 1: Collecting the Data

In this section, you will run the collection content for both modules from the BigFix Console. Exercise 2 will have you run it through the Workbench.

Client Performance Counter Analysis

In this exercise, you will simply enable an analysis that gathers various Client performance properties for use later by the Collect Client Performance Counter Analysis Data (2.0) task.

1. Switch to the BigFix Server virtual machine – **BESFNDWINROOT** – and login to the machine using the following local administrator credentials:

   Username: Administrator

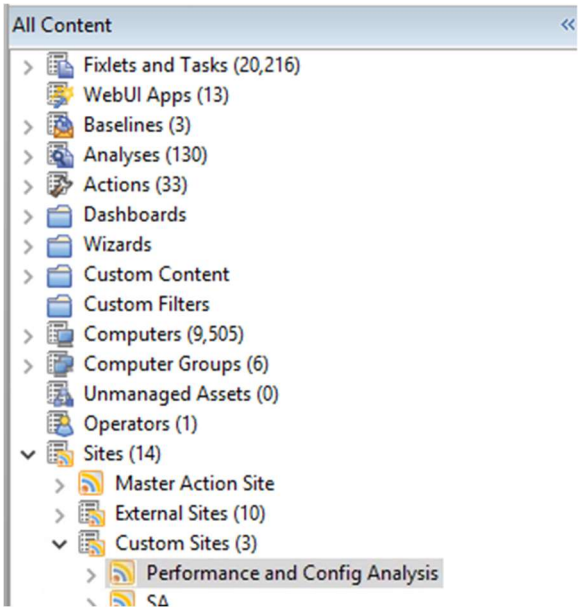
   Password: bigfixrocks

2. Launch the BigFix console and login with the following Master Operator credentials:

   Username: adminmo

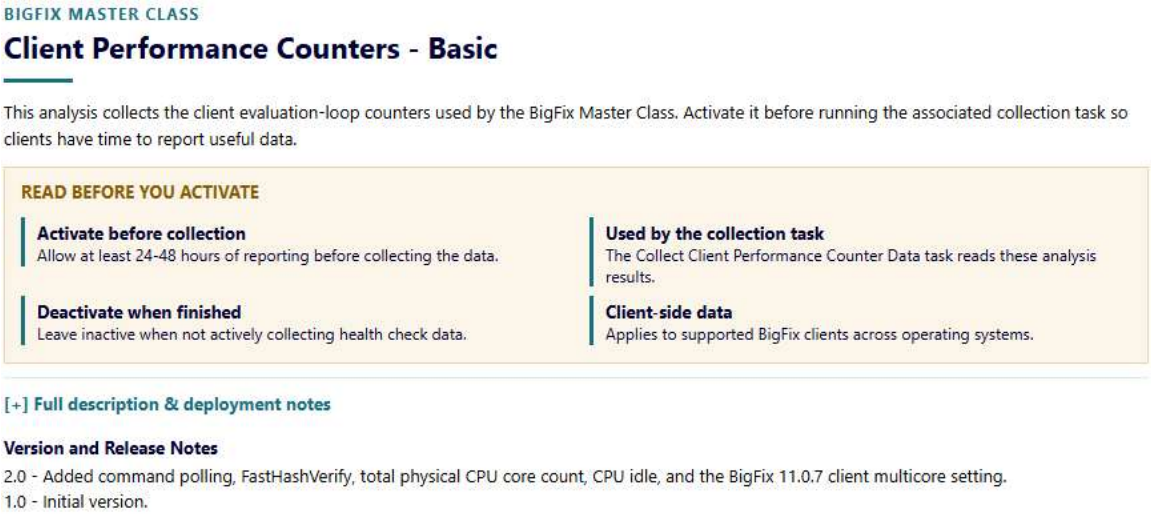
   Password: B1gfixrocks

3. Select the **All Content** domain. Expand **Sites > Custom Sites**. Select the **Performance and Config Analysis** site.

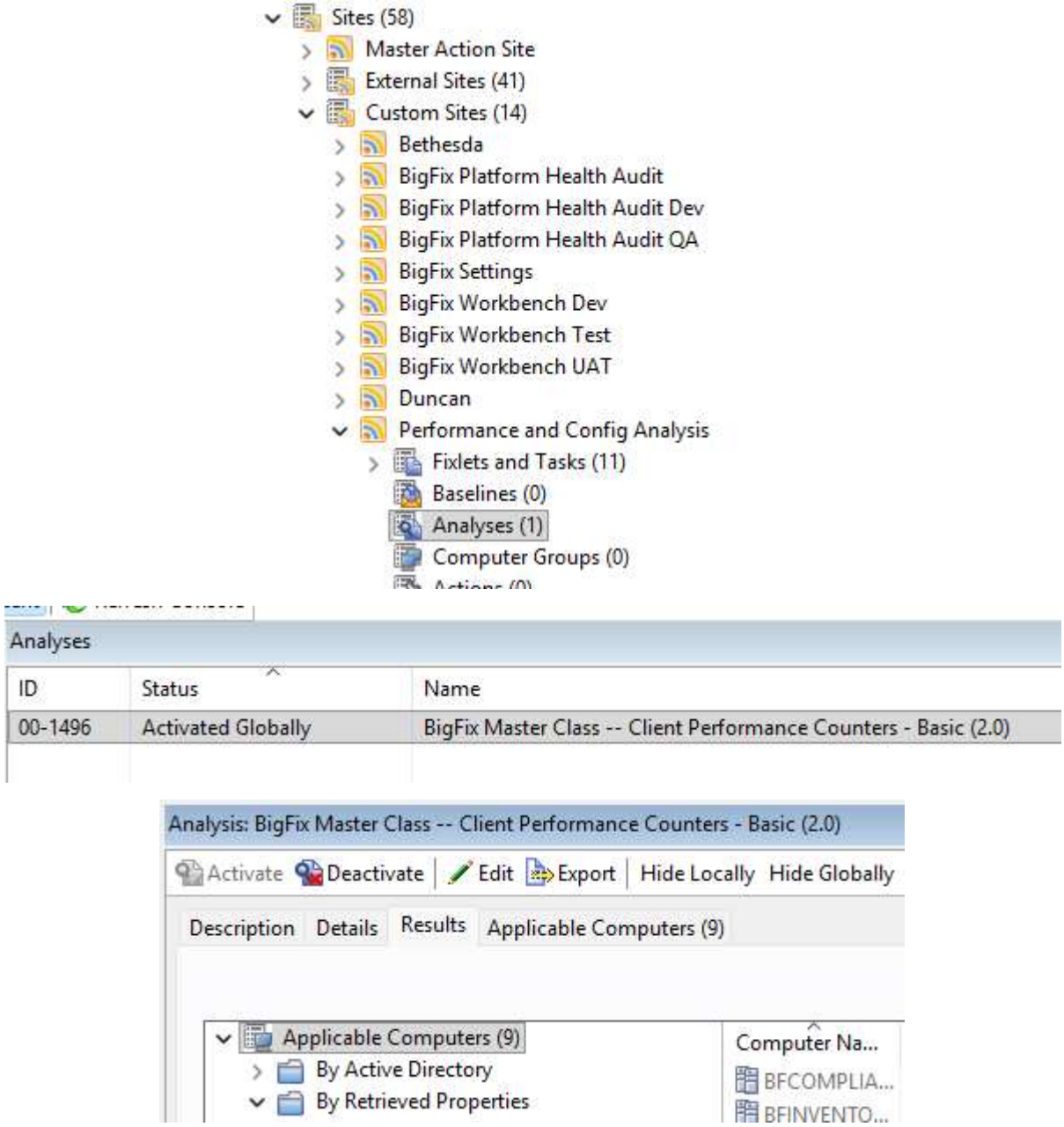


4. Within that site, select **Analyses** then the **BigFix Master Class -- Client Performance Counters - Basic (2.0)** analysis. Review the **Description** to get familiar with what it does.

   NOTE: Be sure to click the '+' sign next to Full Description & Deployment Notes to view all of the details.



5. Finally, in the top navigation pane, right-click on the **BigFix Master Class -- Client Performance Counters - Basic (2.0)** analysis and choose *Activate*. The status changes to *"Activated Globally*" (Globally because you are logged on as a Master Operator right now) and the **Results** tab will appear. Data will then start to show up gradually.



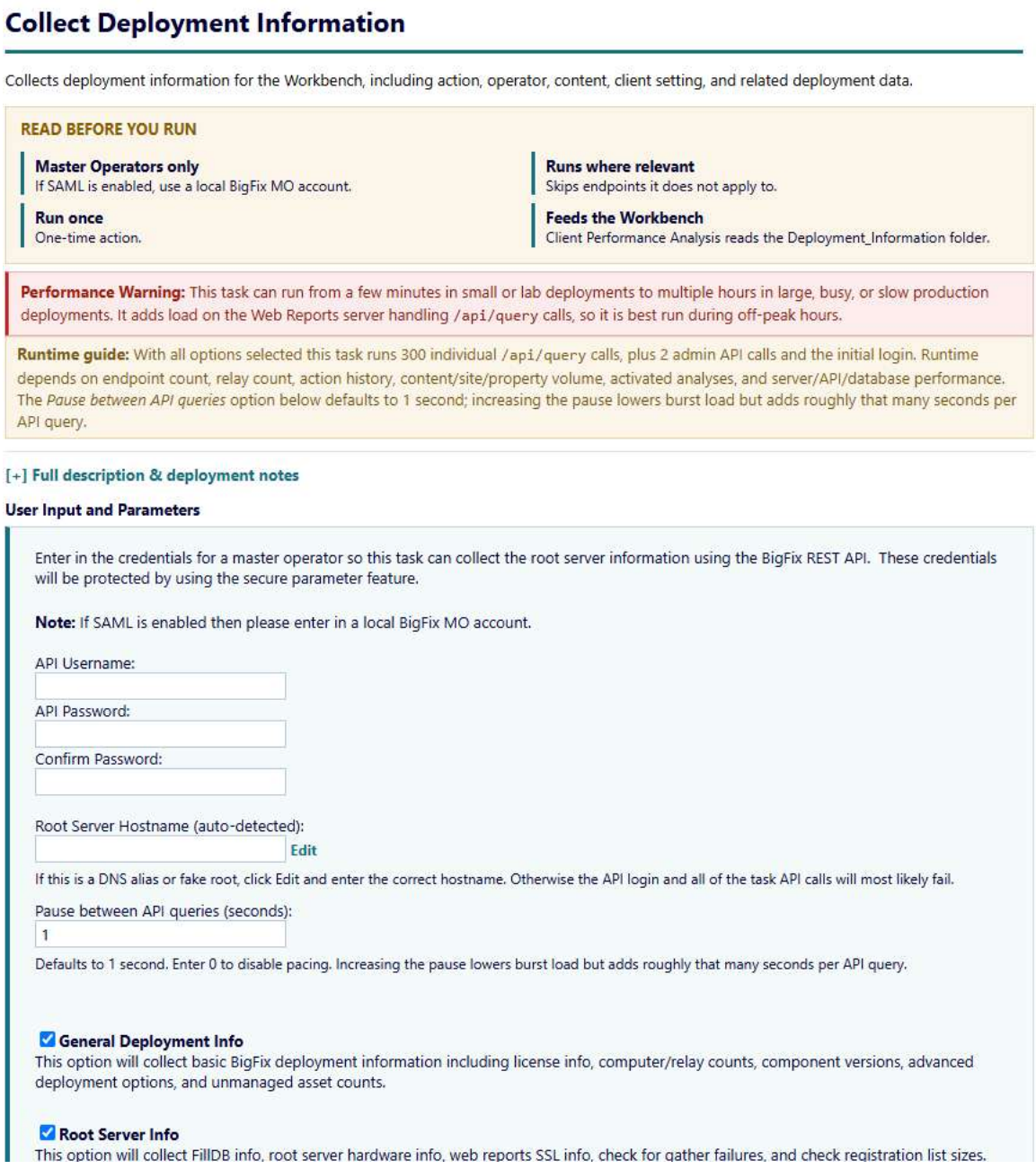
NOTE: In typical troubleshooting / health check scenarios, the analysis should be activated for 24 to 48 hours before you collect data. This is to ensure that you get as many computers reporting in as possible.

Deployment Information

Now you will run the task to collect deployment details. This is fairly large task and it retrieves a lot of useful information. In addition to the specific use case here of Client Performance analysis, this task is also used as a part of BigFix Health Checks, etc. and is a nice tool to have around.

1. Still logged in to the BigFix Console, go to the **Performance and Config Analysis** site, select **Fixlets and Tasks** and then **BigFix Master Class -- Collect Deployment Information (2.0).** The task description opens in the work area. Spend a minute or so reviewing the details in the **Description** tab. Also take a minute to review the Actionscript and note the various parameters available to control files names, etc.

   NOTE: Be sure to click the '+' sign next to Full Description & Deployment Notes to view all of the details.



2. Enter the console user name in *API Username* then enter the console password in *API Password and Confirm Password* fields. Leave Root Server Hostname as it is. Change it only for a DNS alias or fake root.

   Username: adminmo

Password: B1gfixrocks

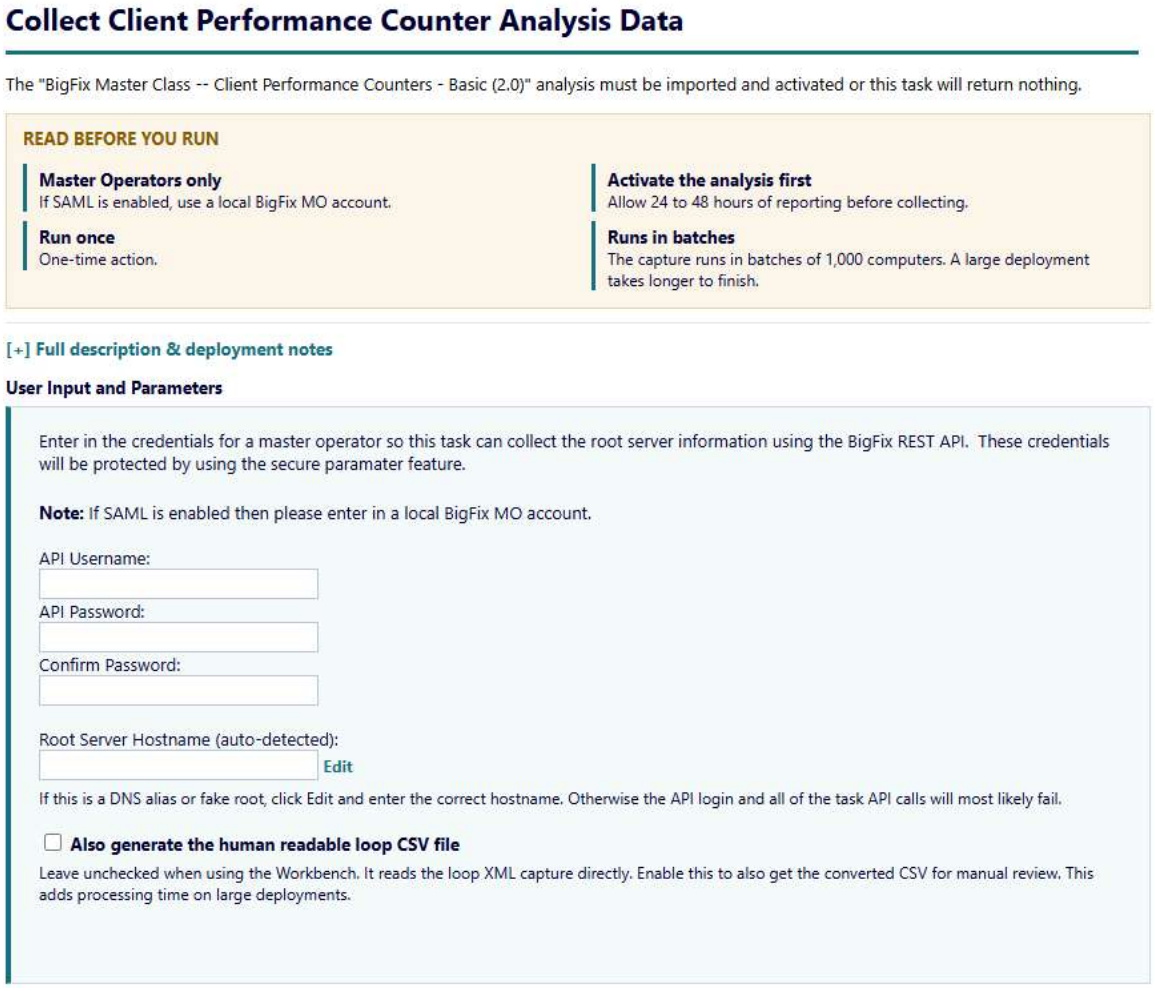
3. Click **Take Action**. On the Target tab select the **BESFNDWINROOT**. Click **OK**.
4. The action starts. You can review the BigFix client logs to see the action executing but, in the interest of time, do not wait for it to complete as this task queries a lot of information and can take a while to run. Continue on to the next section.

Collect Client Performance Counter Analysis Data

In this section, you will run an action that pulls the results from the analysis you activated in the Client Performance Counter analysis exercise. That data is retrieved and formatted in such a way as to make it useful for the Workbench's Client Performance module.

1. Select **BigFix Master Class -- Collect Client Performance Counter Analysis Data (2.0).** The task description opens in the work area. Spend a minute or so reviewing the details in the **Description** tab.

   NOTE: Be sure to click the '+' sign next to Full Description & Deployment Notes to view all of the details.

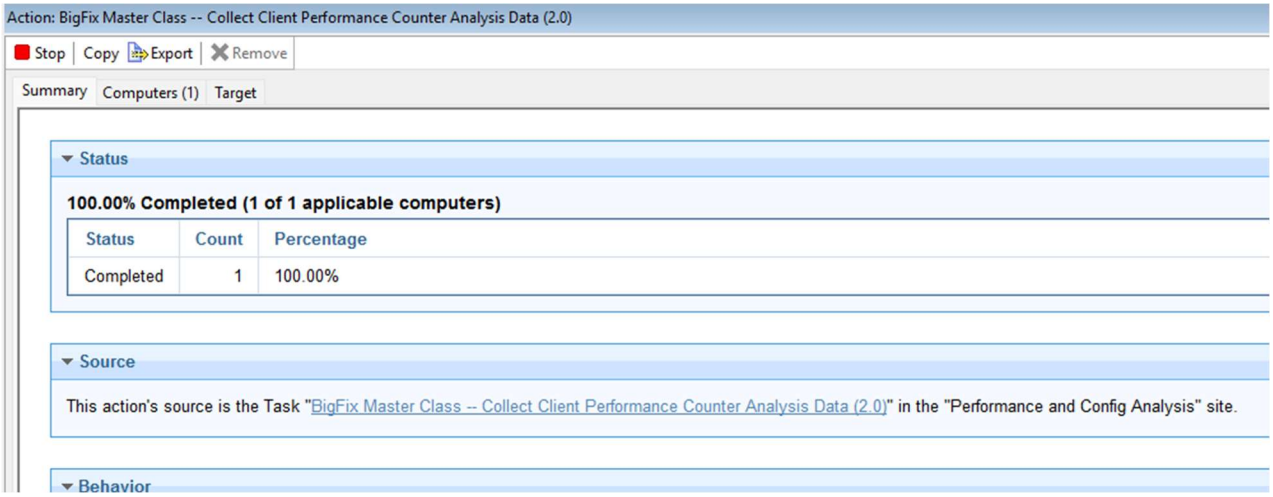


2. Enter the console user name in *API Username*. Enter the console password in *API Password and Confirm Password*. Leave the Root Server Hostname as it is.

   Username: adminmo

   Password: B1gfixrocks

3. Leave the "Also generate the human readable loop CSV file" option cleared.
4. Click **Take Action**. On the Target tab select BESFNDWINROOT. Click **OK**.
5. Wait for the action status to change to *Completed*. This should only take a few moments since there are only a small group of machines in the lab.



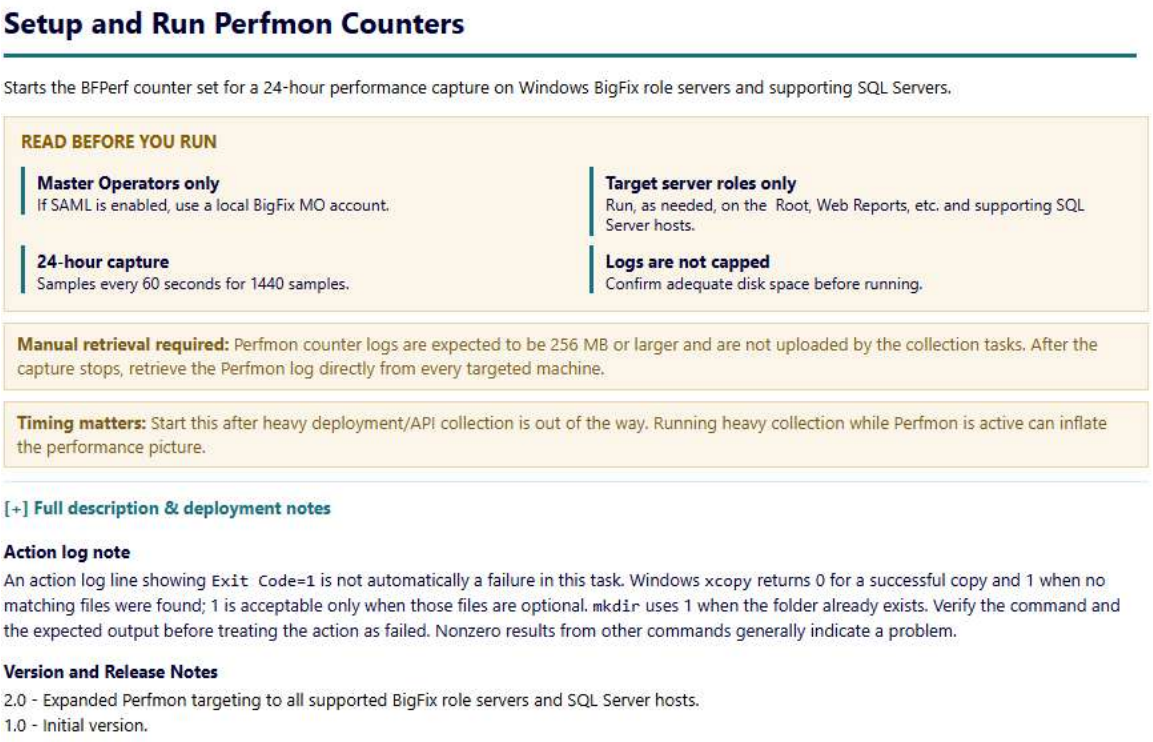
6. You can view the data by opening the **"C:\BigFix_Master_Class\Platform\Results\Client_Loop\"** folder.

Perfmon Counters

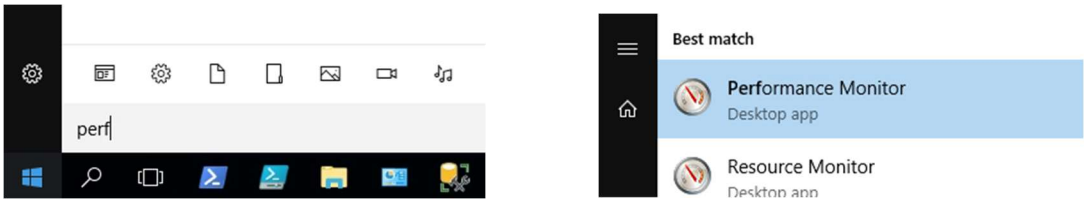
In this lab you will run an action that configures and runs a custom Windows Perfmon data collector / counter set. The resulting file (.blg) is then used as the input for the Workbench's Perfmon Analysis module.

1. Select **BigFix Master Class -- Setup and Run Perfmon Counters (2.0).** The task description opens in the work area. Spend a minute or so reviewing the details in the **Description** tab.

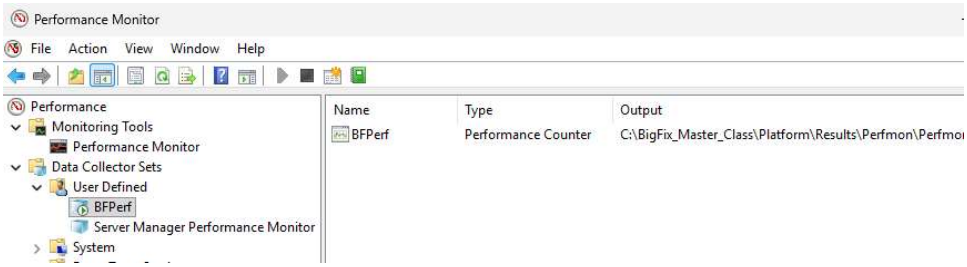
   NOTE: Be sure to click the '+' sign next to Full Description & Deployment Notes to view all of the details.



2. Click **Take Action**. On the Target tab select BESFNDWINROOT. Click **OK**.
3. Wait for the action status to change to *Completed*.
4. Now let's verify the new "BFPerf" collector set is present and active in Performance Monitor.
5. Using the Windows Start menu, type "perf" (for performance) and then open the Performance Monitor application.



6. In the Performance Monitor app, use the navigation tree and go to **Data Collector Sets > User Defined**. You should see **"BFPerf"** under that folder and it has a small green "play button" icon. This confirms that the new Perfmon counter set has been properly configured and is running.



NOTE: The capture takes a sample every 60 seconds and stops on its own after 24 hours. You do not need to stop it. The Workbench uses the sample capture in Exercise 2.

7. You can view the data by opening the **"C:\BigFix_Master_Class\Platform\Results\Perfmon\"** folder. Note that active Perfmon counter files will report a 0-size file until Perfmon capture is stopped.

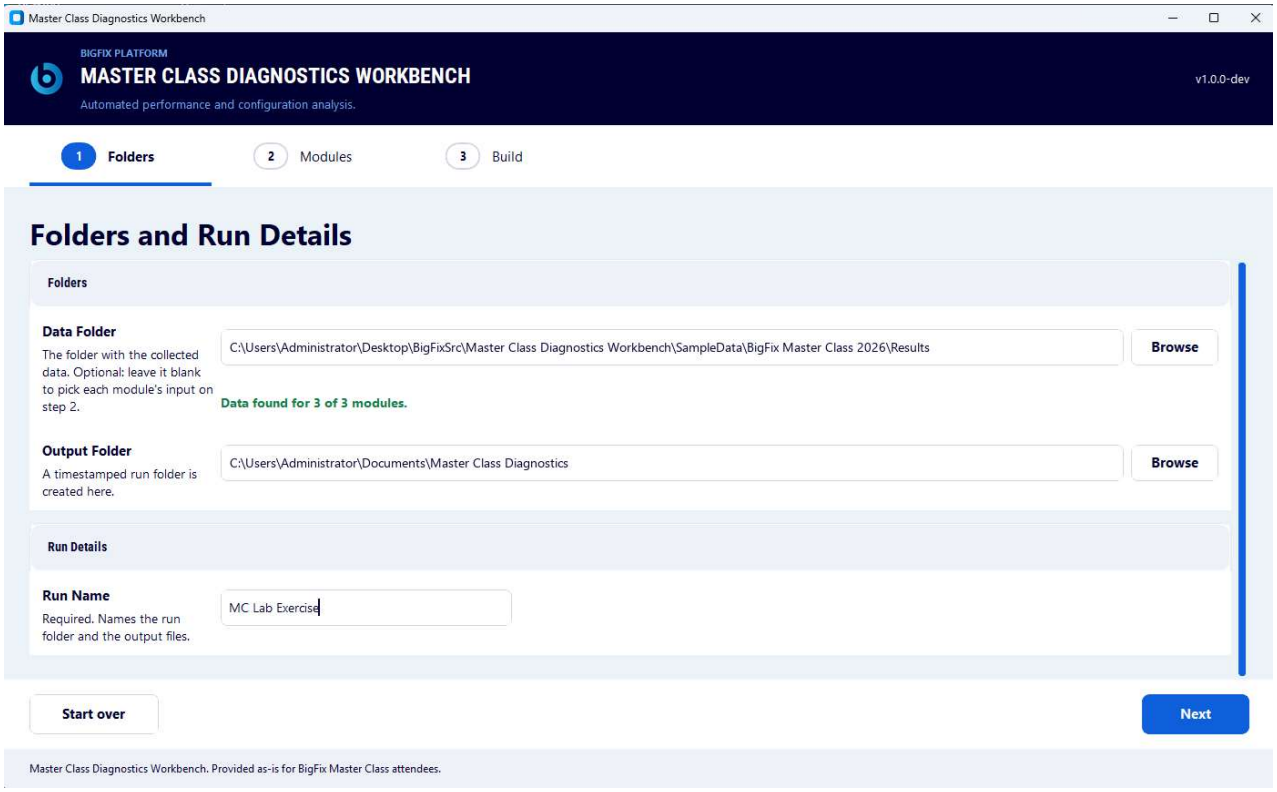
You have now successfully completed Exercise 1.

Exercise 2: Run the Workbench

In this exercise you will run the two Workbench modules against the lab sample data.

Step 1: Folders and Run Details

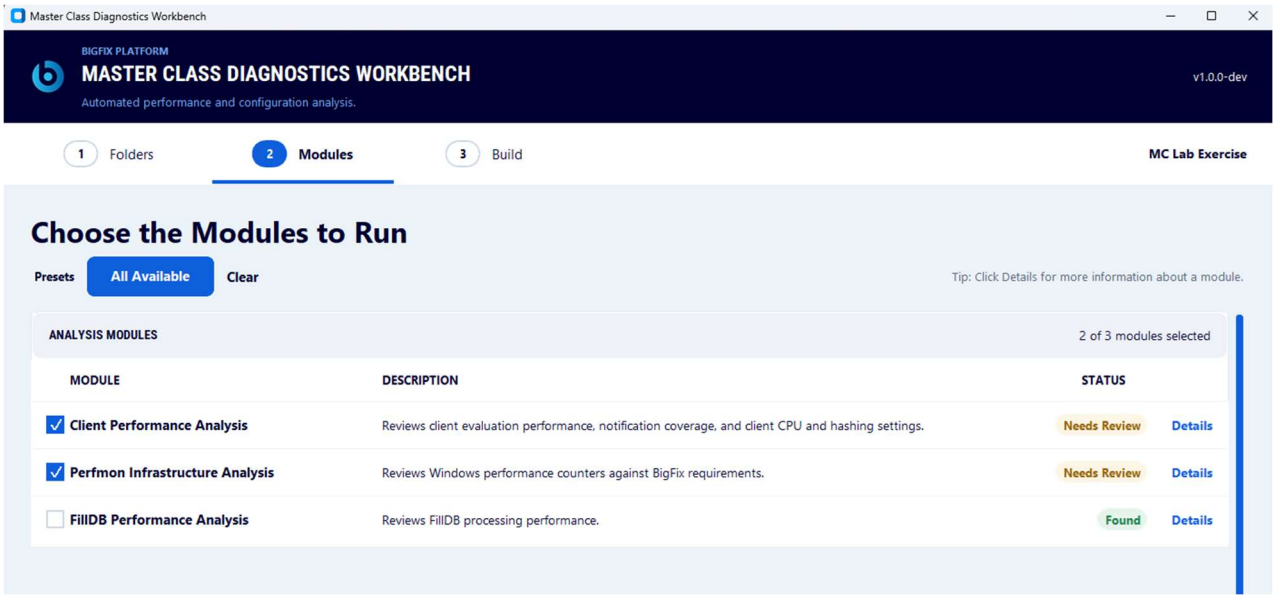
1. On the BigFix Server open **C:\Users\Administrator\Desktop\BigFixSrc\Master Class Diagnostics Workbench\Launchers**. Double-click **Run Diagnostics Workbench.cmd**. The Workbench opens on step 1.
2. Click **Browse** next to *Data Folder*. Select the **C:\Users\Administrator\Desktop\BigFixSrc\Master Class Diagnostics Workbench\SampleData\BigFix Master Class 2026\Result** folder. The Workbench now reports *"Data Found for 3 of 3 modules".*
3. Leave the *Output Folder* at **C:\Users\Administrator\Documents\Master Class Diagnostics**. A timestamped run folder is created there.
4. Enter a **Run Name** such as *MC Lab Exercise*. Any name works. The run folder and every output file use it.



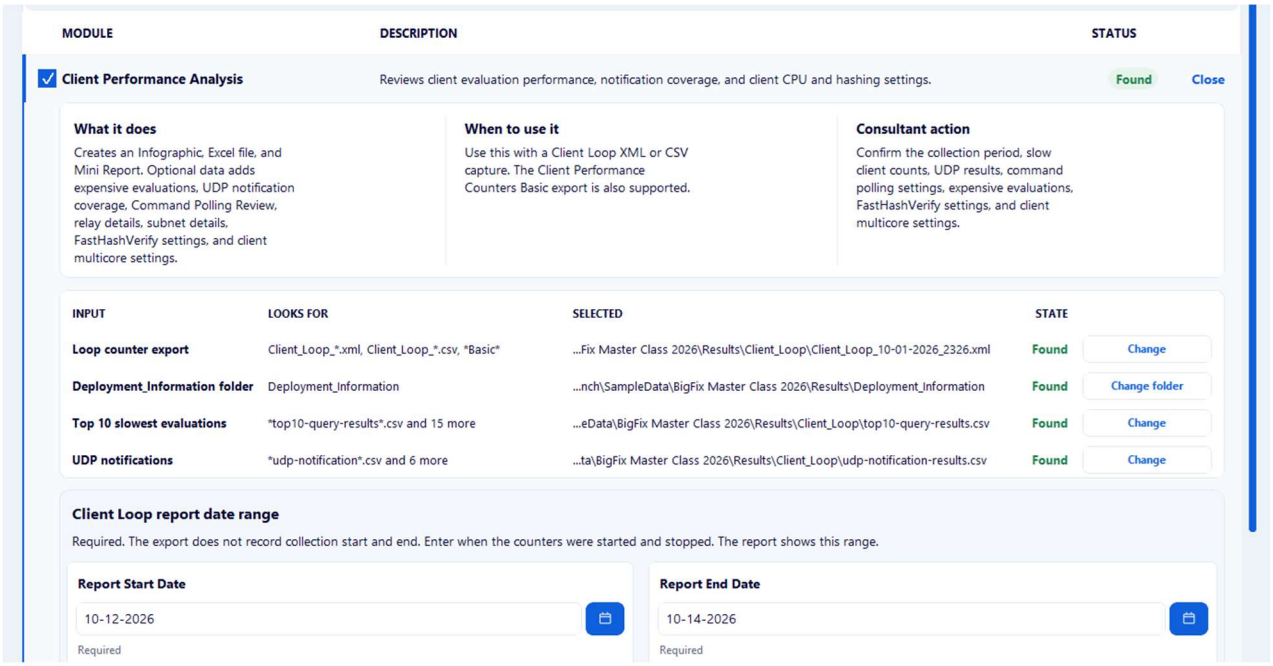
5. Click **Next**. The *Choose the Modules to Run* page opens.

Step 2: Choose the Modules

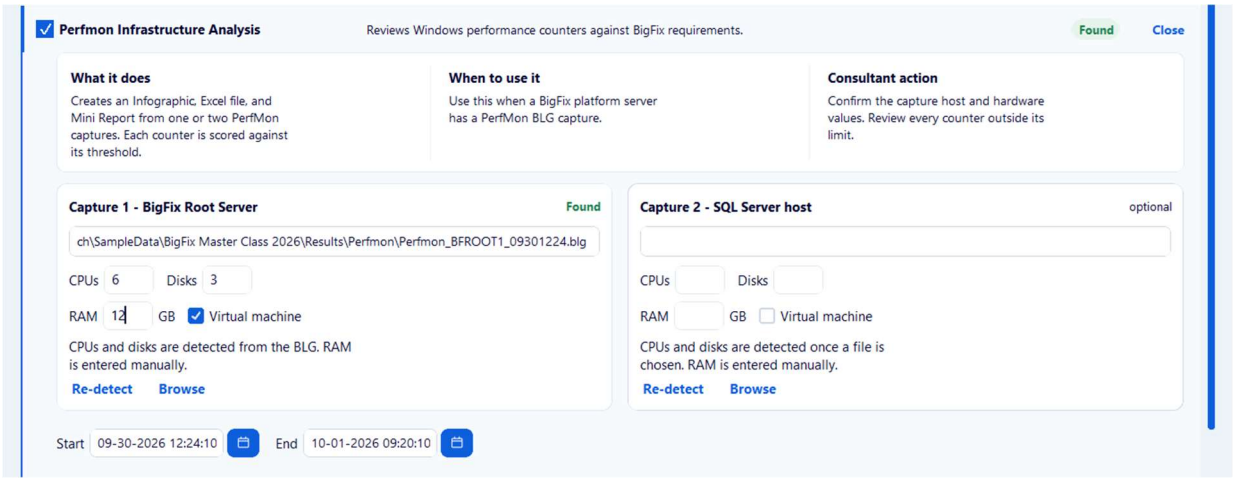
1. The page opens with every module selected. Click **Clear**. Select both **Client Performance Analysis** and **Perfmon Infrastructure Analysis**. Both rows will show *Needs Review* until you fill in their details.



2. Click **Details** on the **Client Performance Analysis** row. Enter a **Report Start Date** and a **Report End Date**. Any dates work. The report shows this range. The row changes to *Found*.



3. Click **Details** on the **Perfmon Infrastructure Analysis** row. In Capture 1 enter **12 in RAM**. Select **Virtual machine**. CPUs and Disks are read from the .blg file capture. The row changes to *Found*.

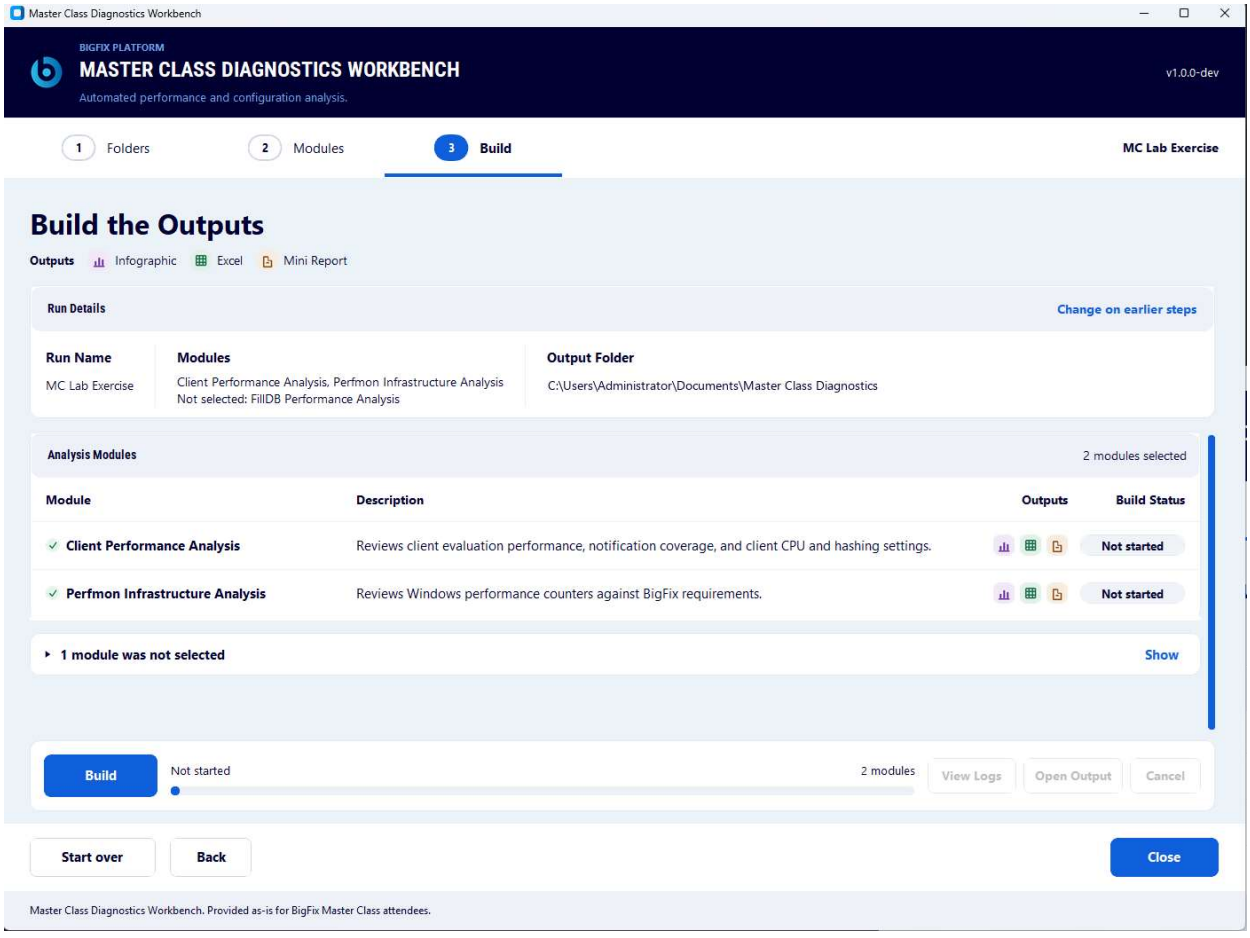


NOTE: RAM is not in the Perfmon capture. Without it the Memory Health counter (PRF-INF-09) is not scored.

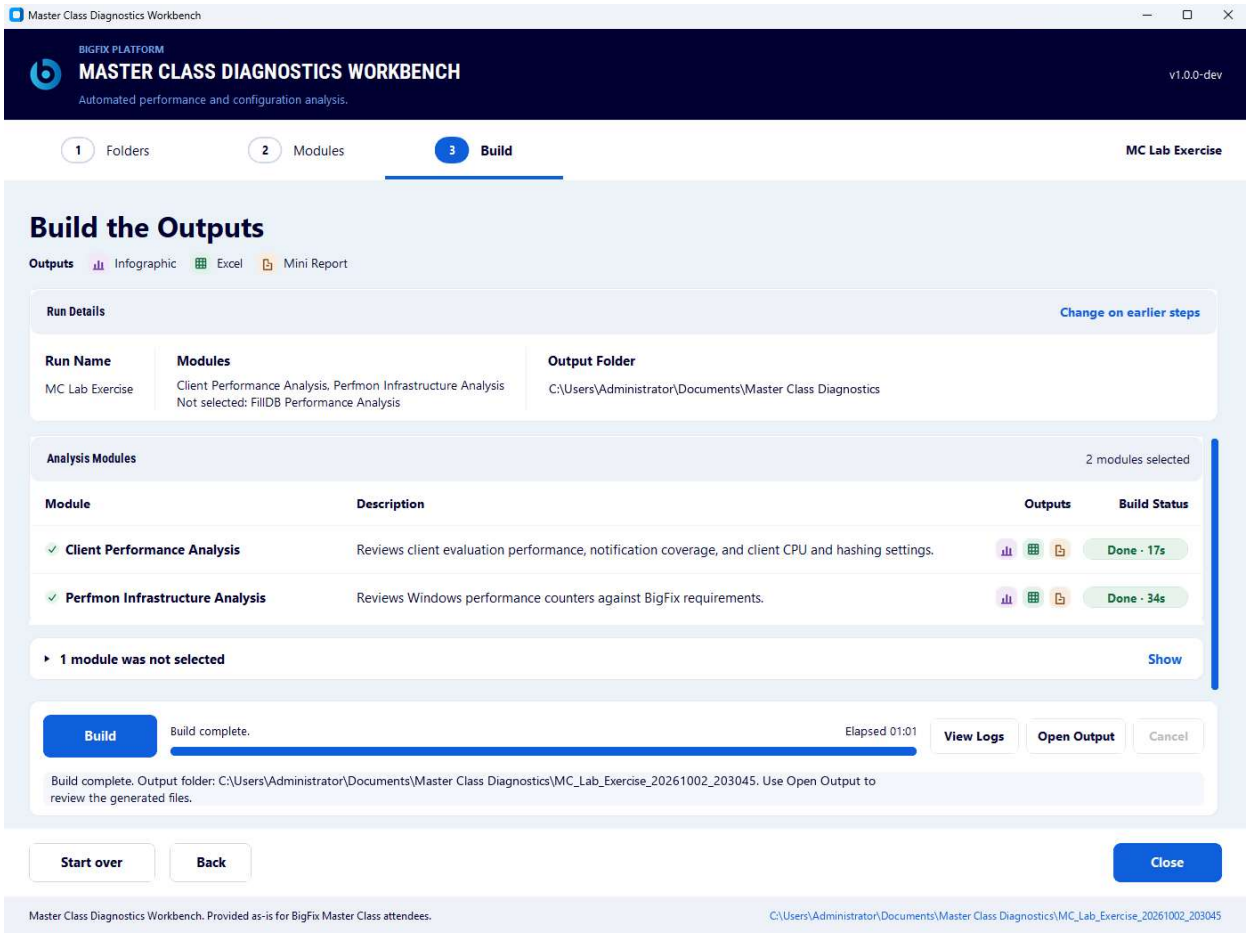
4. Click **Next**. The **Build the Outputs** page opens.

Step 3: Build the Outputs

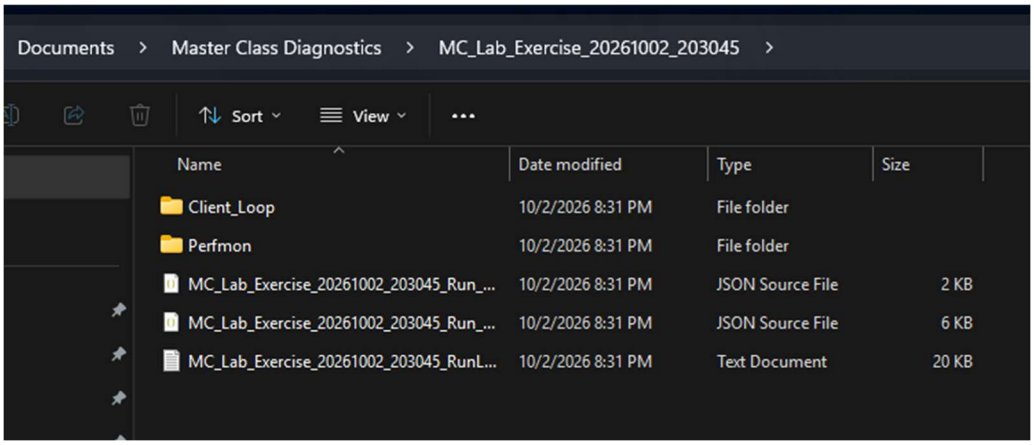
1. Check the **Run Details**. Click **Build**.



2. Wait for the *Build Status* of each module to change from **Not started** to **Running** to **Done**. You can monitor the status of the build in more detail by clicking on **View Logs**.



3. Click **Open Output**. The run folder opens. It holds a folder for each module along with the run logs.

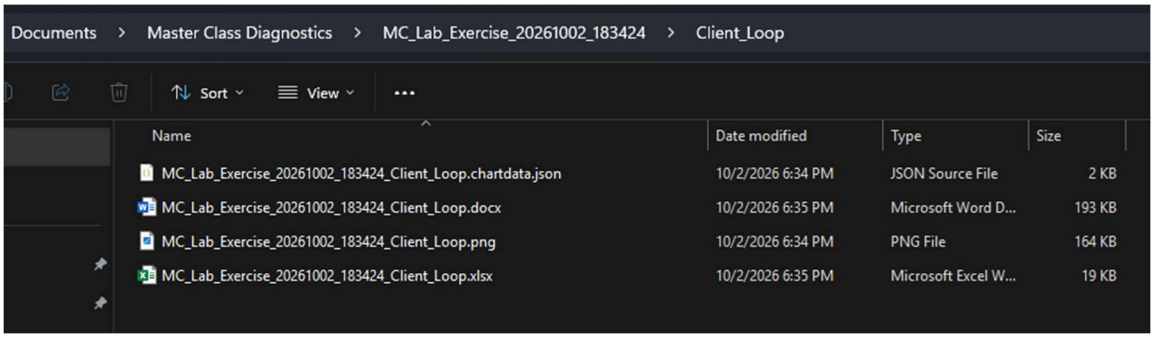


You have now successfully completed Exercise 2.

Exercise 3: Read the Client Performance Output

In this exercise, you will read the Client Performance outputs. Each Workbench module writes a Mini Report, an Infographic and an Excel workbook.

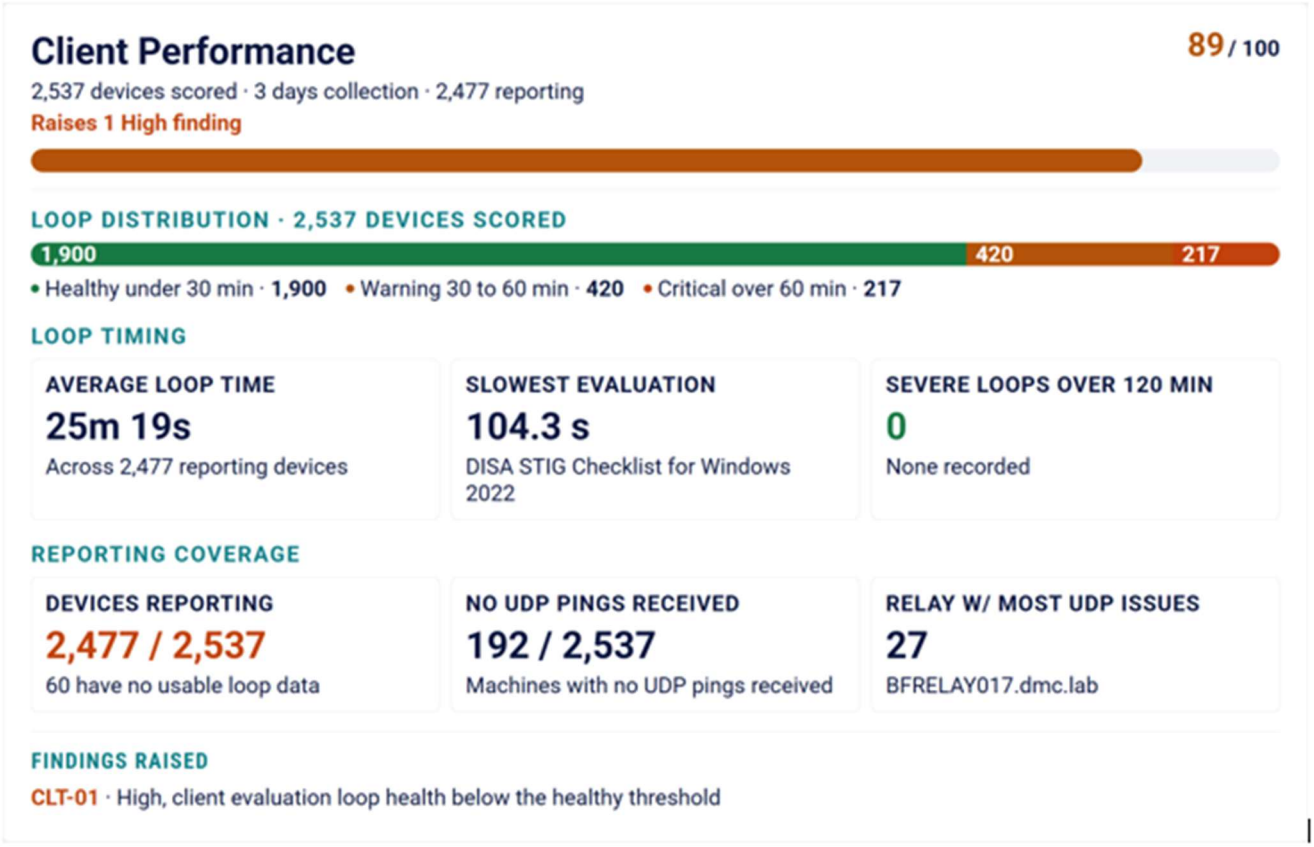
1. Open the Client_Loop folder at **C:\Users\Administrator\Documents\Master Class Diagnostics\<run name>_<date>_<time>\Client_Loop**



2. Open and review the Mini Report .docx as well as the Excel workbook to see what data is available. Discussion questions will be presented to class.

Client Performance Analysis

This report covers 2,537 BigFix Clients from 10-01-2026 to 10-03-2026. It was generated on 10/6/2026 23:54:32. The Client Performance workbook (Excel file) holds the per-device data behind every table.



Reading the Infographic

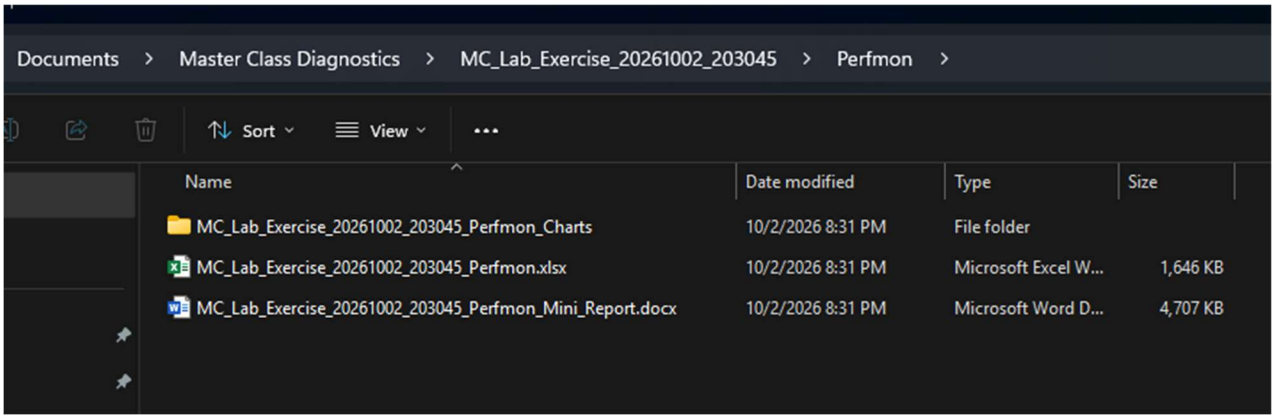
Infographic tile	Value	What it means
Health score	89 / 100	Average of each device's band points. See Loop Health.
Loop distribution	1,900 · 420 · 217	Devices averaging under 30, 30 to 60, and over 60 minutes per loop.
Average loop time	25m 19s	Mean of every device's average loop time.
Slowest evaluation	104.3 s	Top entry in Top 10 Most Expensive Evaluations.

You have now successfully completed Exercise 3.

Exercise 4: Read the Perfmon Analysis Output

In this exercise, you will review the Perfmon analysis outputs. Each counter is scored against its target in the NFR guide. The NFR guide standards and thresholds are listed below for your convenience.

1. Open the Perfmon folder **C:\Users\Administrator\Documents\Master Class Diagnostics\<run name>_<date>_<time>\Perfmon**



2. Open and review the Mini Report .docx as well as the Excel workbook to see what data is available. Discussion questions will be presented to class.

The link to the source document is [BigFix Non-Functional Requirements: A Checklist Approach, v11.x.4.]

ID	Area	Performance counter	Standard
PRF-INF-01	Storage Queue Depth	**Physical Disk: Current Disk Queue Length**	≤ 1 per physical device
PRF-INF-02	Storage Latency	**Physical Disk: Avg. Disk sec/Transfer**	≤ 1 ms per physical device
PRF-INF-03	Storage Operations	**Physical Disk: Disk Transfers/sec**	No saturation under load. DBMS storage supports 5,000 transfers per second.
PRF-INF-04	CPU Utilization	**Processor: % Processor Time**	Virtual: 50% to 80%. Physical: ≤ 80%.
PRF-INF-05	CPU Latency	**System: Processor Queue Length**	≤ 4
PRF-INF-06	CPU Privileged Time	**Processor: % Privileged Time**	≤ 10% of aggregate CPU
PRF-INF-07	Network Health	**Network Interface: Packets Outbound Discarded, Packets Outbound Errors, Packets Received Discarded, Packets Received Errors**	All counters approach zero per interface

ID	Area	Performance counter	Standard
PRF-INF-09	Memory Health	**Paging File: % Usage Peak**	≤ 5% of aggregate physical memory

Perfmon Infrastructure Analysis

PerfMon Capture 1

BFROOT1

PerfMon Counters

62 / 100

8 checks against thresholds · values are capture averages

CHECKS AGAINST THRESHOLDS · 8 CHECKS

STORAGE QUEUE DEPTH	STORAGE LATENCY	STORAGE OPERATIONS	CPU PRIVILEGED TIME
0.25	2.61 ms	77.99	28.97%
Limit 1.	Limit 1 ms.	Capacity near 5,000 per second.	Limit 10% of CPU time.

CPU UTILIZATION	CPU LATENCY	NETWORK HEALTH	MEMORY HEALTH
53.87%	1.14	0	24.2%
50-80% (VM).	Limit 4.	Errors should approach zero.	Limit 5% (physical memory).

FINDINGS RAISED

3 total · 3 critical

IDs PRF-INF-02, PRF-INF-06, PRF-INF-09

How This Score Is Calculated

The PerfMon Infrastructure Score is based on how each monitored counter area compares to its applicable threshold during the capture window:

- Pass: 100 points

- Advisory: 55 points

- Cannot Verify: 25 points

- Fail: 0 points

Formula: Pass% x 100 + Advisory% x 55 + Cannot Verify% x 25 + Fail% x 0

The counter-area status is based on the average observed value, so the score reflects sustained behavior rather than a single short spike.

Score bands: 85-100 Healthy, 60-84 Needs Attention, 0-59 Critical.

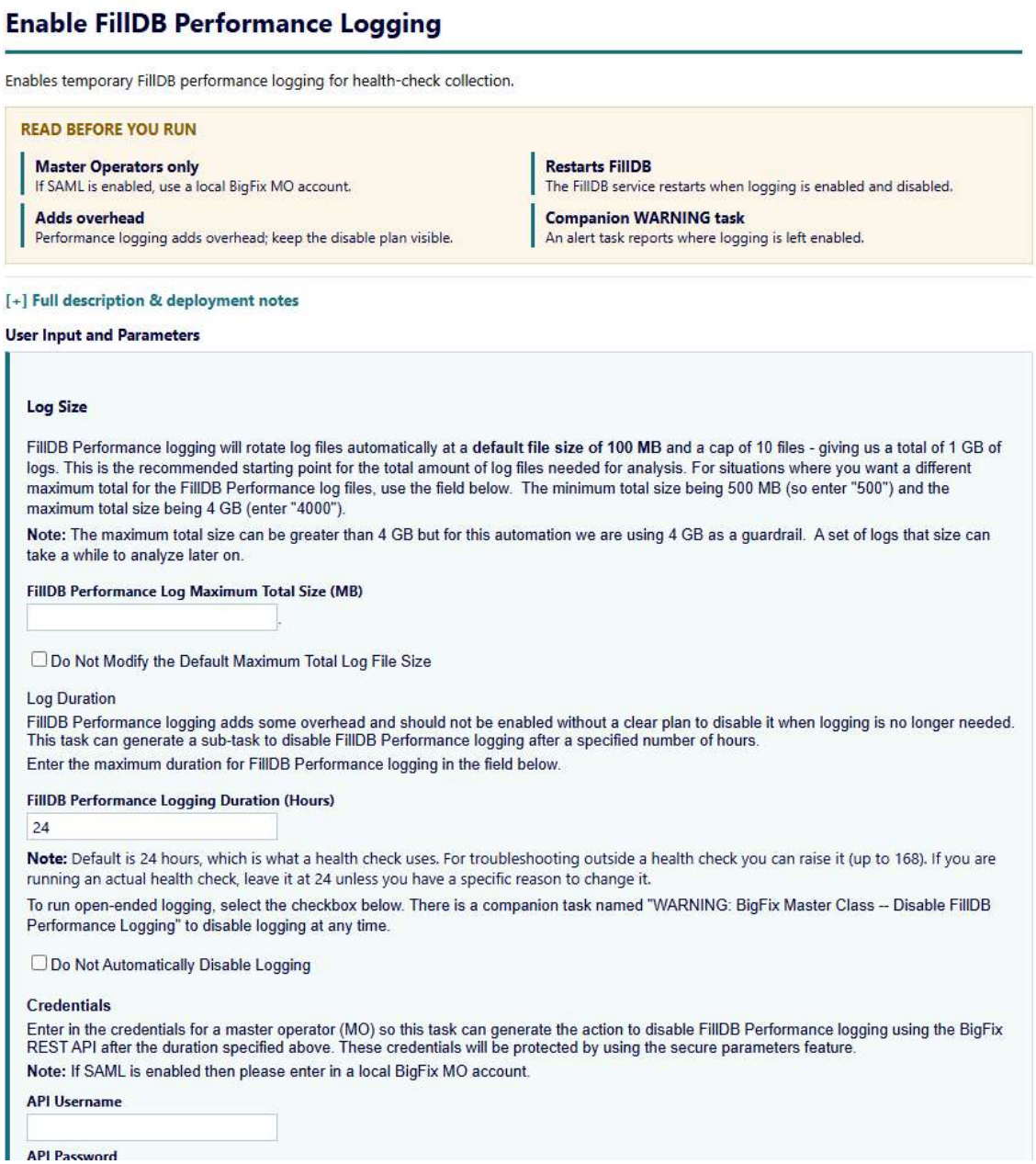
You have now successfully completed Exercise 4.

Appendix A: FillDB Performance (Bonus Round)

FillDB is not part of the live lab. Run it if time permits or after the session. The sample data includes a FillDB analyzer CSV, so you can run the module without collecting logs. FillDB Performance Analysis reviews how quickly the root server processes client reports.

Collect the logs

1. In the Performance and Config Analysis site select **BigFix Master Class -- Enable FillDB Performance Logging (2.0)**.



2. Enter the console user name in *API Username*. Enter the console password in *API Password and Confirm Password*. Keep the default duration.

   Username: adminmo

   Password: B1gfixrocks

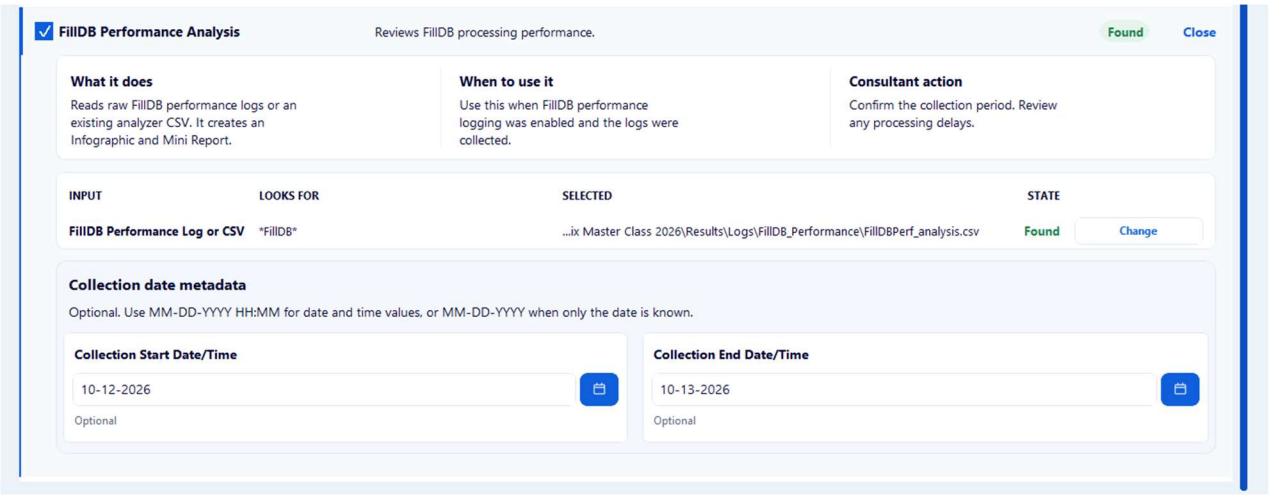
3. Click **Take Action**. On the **Target** tab select BESFNDWINROOT. Click **OK**.

4. Logging stops on its own after the default 24-hour duration. If you selected *Do Not Automatically Disable Logging*, run **WARNING: BigFix Master Class -- FillDB Performance Logging is Enabled (2.0)** to stop it.

5. FillDB writes FillDBPerf.log and its rollover files to **C:\BigFix_Master_Class\Platform\Results\Logs\FillDB_Performance** on the root server.

Run the module

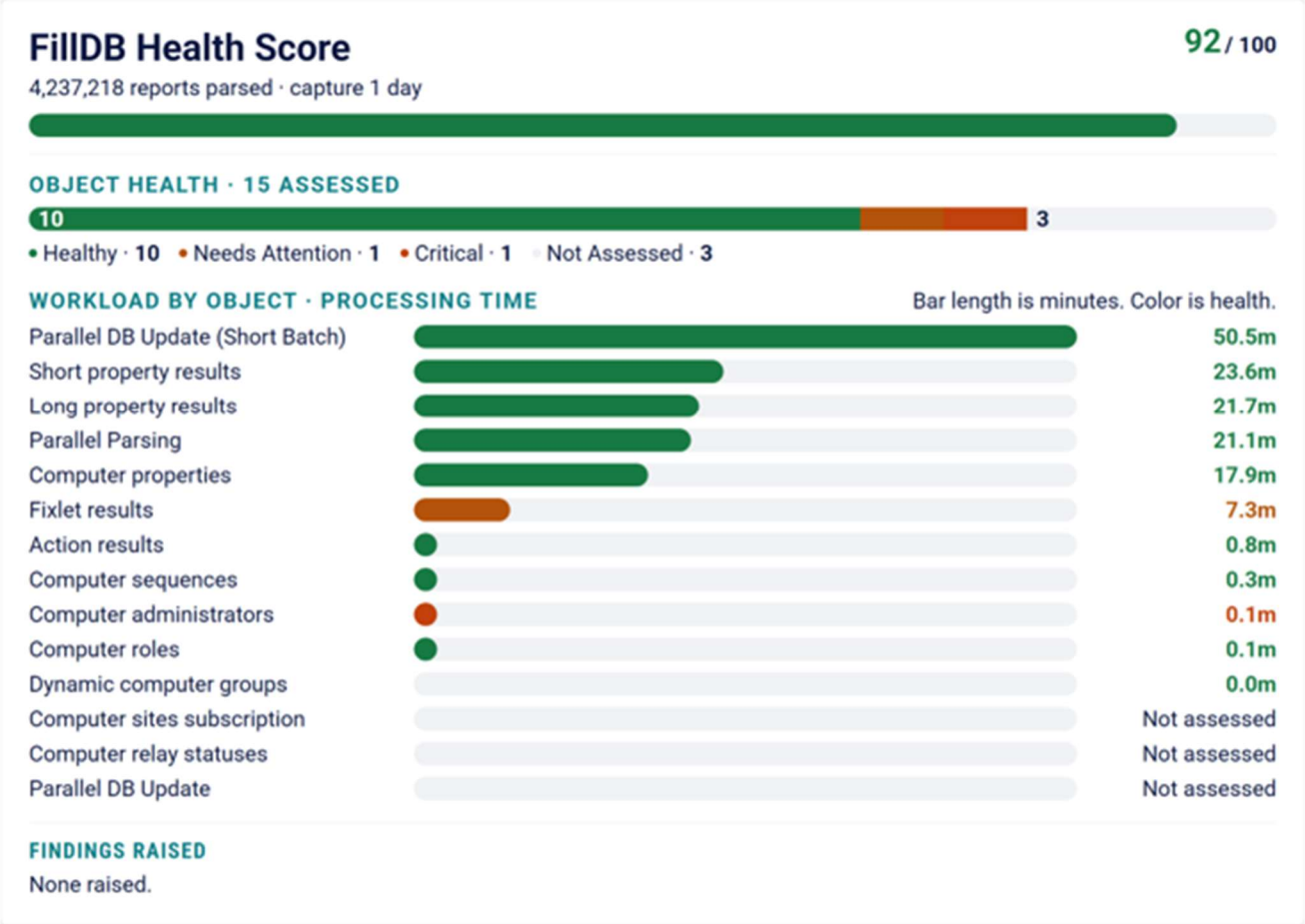
1. Run the Workbench. On step 2 click **Clear** and select **FillDB Performance Analysis**. Click **Details** to check the log. The collection dates are optional.



2. Click **Next** and click **Build**.

3. Open the **FillDB_Performance** folder in the run folder. Open and review the Mini Report .docx as well as the Excel workbook to see what data is available. Discussion questions will be presented to class.

FillDB Performance Analysis



How This Score Is Calculated

The FillDB Health Score is based on the percentage of assessed FillDB objects in each analyzer health band:

- Great: 100 points

- Good: 67 points

- Poor: 34 points

Formula: Great% x 100 + Good% x 67 + Poor% x 34

This score is separate from the Summary Scorecard category scores; it is a focused technical signal for the FillDB capture.

Rows without a recognized Health Check result are shown for context but are not included in the score.

Score bands: 85-100 Healthy, 60-84 Needs Attention, 0-59 Critical.

You have now successfully completed this exercise.